

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (Electrical Engineering)

November 2012

EE 304 Industrial Instrumentation

Date: 27.11.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What is Thermistor? Explain their different forms of construction. Draw their resistivity versus temperature characteristics and show that they have a very high value of sensitivity. [07]

Q - 2 (a) Explain the construction and working of LVDT in detail. [08]

(b) Explain different methods used for torque measurement. [06]

OR

Q - 2 (a) Enlist the methods used for the measurement of Humidity and explain any two in detail. [07]

(b) Explain the difference between Measurement and Instrumentation and describe the static and dynamic characteristics of instrumentation system. [07]

Q - 3 Explain any two: [14]

[1] Radiation Pyrometers

[2] Resistive methods for liquid level measurement

[3] Thermocouple Vacuum Gauge

[4] Classifications of Transducers

SECTION - II

Q - 4 (a) Describe the methods used for measurement of angular velocity and explain any one in detail. [07]

Q - 5 (a) What is Piezo-electric effect? Derive the relationship between output voltage and applied pressure. [08]

(b) A strain gauge having a resistance of 200Ω and gauge factor of 2.5 is connected in series with a ballast resistance of 400Ω across a 24 volt supply. Determine the change in output voltage when stress of 140 MN/m^2 is applied. Take modulus of elasticity is 200 GN/m^2 . [06]

OR

[P.T.O.]

Q - 5 (a) Explain the method of calibration of strain gauges. [07]

(b) Derive the equation of output voltage of voltage sensitive bridge used for measurement of strain. [07]

Q - 6 Attempt any two: [14]

[1] Hot wire anemometer

[2] Turbine Flow meters

[3] Strip Chart Recorders

[4] Digital Recorders
